```
import math

# Part A: Machine Initialization
machines = ['A', 'B', 'C']
N_play = 10

# Initialize tracking tables
N = {m: 0 for m in machines}
R = {m: [] for m in machines}
Q = {m: 0.0 for m in machines}
UCB = {m: float('inf') for m in machines}

print("--- Multi-Armed Bandit: UCB Algorithm ---")

for t in range(1, N_play + 1):
    print(f"\n================ Play {t} ================")

    # Part C: Calculate UCB and select machine with highest value
    for m in machines:
        if N[m] == 0:
            UCB[m] = float('inf')
        else:
            UCB[m] = Q[m] + math.sqrt((2 * math.log(t)) / N[m])
        print(f"Machine {m}: N = {N[m]}, Q = {Q[m]:.3f}, UCB = {UCB[m]:.3f}")

    selected_machine = max(machines, key=lambda m: UCB[m])
    print(f"\nSelected Machine: {selected_machine}")

    # Part D: Reward Collection
    while True:
        try:
            reward = float(input(f"Enter reward for Machine {selected_machine}: "))
            break
        except ValueError:
            print("Invalid input. Enter a valid number.")

    # Part E: Update Statistics and Incremental Q-Value
    N[selected_machine] += 1
    R[selected_machine].append(reward)
    Q[selected_machine] += (1 / N[selected_machine]) * (reward - Q[selected_machine])

# Part F: Display Results
print("\n" + "=" * 55)
print("             FINAL EXPERIMENT RESULTS")
print("=" * 55)
print(f"{'Machine':<10}{'N (Plays)':<12}{'Q-Value':<12}{'Final UCB':<12}{'Rewards History'}")
print("-" * 55)

total_reward = 0
```

```
for m in machines:
    total_reward += sum(R[m])
    ucb_str = f"{UCB[m]:.3f}" if UCB[m] != float('inf') else "inf"
    print(f"{m:<10}{N[m]:<12}{Q[m]:<12.3f}{ucb_str:<12}{R[m]}")

print("-" * 55)
print(f"Total Cumulative Reward Obtained: {total_reward}")
```

```
================ Play 5 ================
Machine A: N = 1, Q = 3.000, UCB = 4.794
Machine B: N = 2, Q = 5.500, UCB = 6.769
Machine C: N = 1, Q = 4.000, UCB = 5.794

Selected Machine: B
Enter reward for Machine B: 6

================ Play 6 ================
Machine A: N = 1, Q = 3.000, UCB = 4.893
Machine B: N = 3, Q = 5.667, UCB = 6.760
Machine C: N = 1, Q = 4.000, UCB = 5.893

Selected Machine: B
Enter reward for Machine B: 8

================ Play 7 ================
Machine A: N = 1, Q = 3.000, UCB = 4.973
Machine B: N = 4, Q = 6.250, UCB = 7.236
```

```
A         1           3.000       5.146       [3.0]
B         8           6.375       6.954       [4.0, 7.0, 6.0, 8.0, 6.0, 7.0, 5.0, 8.0]
C         1           4.000       6.146       [4.0]
-----------------------------------------------------
Total Cumulative Reward Obtained: 58.0
```